Problems that are NOT PSETs

Readings:

- [ ] Section 7.1-7.4, Section 7.5.1, and Section 7.7
- [ ] Section 5.4 and Section 6.1-6.6. You may skip Proposition 6.1 and Example 6.8 on p.385-386
- [ ] Section 5.2.1-5.2.3, 5.3.1-5.3.2, 5.3.4
- [ ] Section 4.5, 4.8

Chapter 5 Exponential Distribution and the Poisson Process:

- Properties of the Exponential Distribution (measure time between events)
    - Memoryless
    - P(x1<x2) = lambda 1 / (lambda 1 + lambda 2)
- The Poisson Process (counting process (every poisson process is a counting process), Interarrival and Waiting Time Distributions, properties of poisson; Conditional Distribution of the Arrival Times)
    - N(t): larger than 0 & integer & never decrease -> this is just for counting process
    - To qualify as a poisson process, N(0) = 0, independent increments,
    - $$\text{(iii)}\quad P(N(t+h)-N(t)=1)=\lambda h+o(h)$$ $$\text{(iv)}\quad P(N(t+h)-N(t)\geqslant 2)=o(h)$$
    - $$P\{N(t+h)-N(t)=0\}=1-\lambda h+o(h)$$ $$P\{N(t+h)-N(t)=1\}=\lambda h+o(h)$$ $$P\{N(t+h)-N(t)\geqslant 2\}=o(h)$$
    -
- Generalizations of the Poisson Process (Nonhomogeneous Poisson Process, and familiarize with lecture notes)

Chapter 6 CTMC:

- Birth and Death Processes
- different type of queueing system (MM1, MMs);
- The Transition Probability Function; little o notation;
- forward and backward;
- Limiting Probabilities;
- time reversibility;
- reversed chain and Computing the Transition Probabilities;

Chapter 7 renewal theory:

- Distribution of N(t);

- renewal reward process;
- Alternating Renewal Processes;
- The Inspection Paradox

Exercises:

- [ ] Example 7.6 on p.438-439
- [ ] Example 7.14, 7.15, 7.16, 7.18, 7.19, 7.20. on p.452-459

Exercise 7.42 on p.475-476 in 11ed.

*42. For an interarrival distribution F having mean μ, we defined the equilibrium distribution of F, denoted F_e, by

$$F_e(x) = \frac{1}{\mu} \int_0^x [1 - F(y)]\, dy$$

(a) Show that if F is an exponential distribution, then $F = F_e$.

(b) If for some constant c,

$$F(x) = \begin{cases} 0, & x < c \\ 1, & x \geqslant c \end{cases}$$

show that F_e is the uniform distribution on $(0, c)$. That is, if interarrival times are identically equal to c, then the equilibrium distribution is the uniform distribution on the interval $(0, c)$.

(c) The city of Berkeley, California, allows for two hours parking at all non-metered locations within one mile of the University of California. Parking officials regularly tour around, passing the same point every two hours. When an official encounters a car he or she marks it with chalk. If the same car is there on the official's return two hours later, then a parking ticket is written. If you park your car in Berkeley and return after three hours, what is the probability you will have received a ticket?

Example 5.24, 5.25 on p.343-345 (p.325-327 in 11ed)

Exercise 5.79 on p.371 (p.352 in 11ed).

distribution of the number of customers that enter the store on a given day?

*79. Suppose that events occur according to a nonhomogeneous Poisson process with intensity function $\lambda(t)$, $t > 0$. Further, suppose that an event that occurs at time s is a type 1 event with probability $p(s)$, $s > 0$. If $N_1(t)$ is the number of type 1 events by time t, what type of process is $\{N_1(t), t \geqslant 0\}$?

Example 6.13, 6.14, 6.15 on p.397-399 (377-379 in 11ed)
Example 5.3 on p.296 (p.281 in 11ed)
Example 5.8 and 5.10 on p.306-309 (p.290-293 in 11ed)
Exercise 5.18 on p.340 and the solutions on p.729

*40. Show that if $\{N_i(t), t \geqslant 0\}$ are independent Poisson processes with rate λ_i, $i = 1, 2$, then $\{N(t), t \geqslant 0\}$ is a Poisson process with rate $\lambda_1 + \lambda_2$ where $N(t) = N_1(t) + N_2(t)$.

41. In Exercise 40 what is the probability that the first event of the combined process is from the N_1 process?

Example 5.18, 5.19, 5.20 on p.328-333 (p.311-316 in 11ed)
Exercise 5.64 on p.368 (p.349 in 11ed).

*64. Suppose that people arrive at a bus stop in accordance with a Poisson process with rate λ. The bus departs at time t. Let X denote the total amount of waiting time of all those who get on the bus at time t. We want to determine $\mathrm{Var}(X)$. Let $N(t)$ denote the number of arrivals by time t.

(a) What is $E[X|N(t)]$?
(b) Argue that $\mathrm{Var}[X|N(t)] = N(t)t^2/12$.
(c) What is $\mathrm{Var}(X)$?

Pset problems:
Exercise 7.12 on p.470 in [IPM11e]

12. Events occur according to a Poisson process with rate λ. Any event that occurs within a time d of the event that immediately preceded it is called a d-event. For instance, if $d = 1$ and events occur at times 2, 2.8, 4, 6, 6.6, . . . , then the events at times 2.8 and 6.6 would be d-events.

(a) At what rate do d-events occur?
(b) What proportion of all events are d-events?

Exercise 7.27

27. A machine consists of two independent components, the ith of which functions for an exponential time with rate λ_i. The machine functions as long as at least one of these components function. (That is, it fails when both components have failed.) When a machine fails, a new machine having both its components working is put into use. A cost K is incurred whenever a machine failure occurs; operating costs at rate c_i per unit time are incurred whenever the machine in use has i working components, $i = 1, 2$. Find the long-run average cost per unit time.

Exercise 7.38 on p.502

38. A truck driver regularly drives round trips from A to B and then back to A. Each time he drives from A to B, he drives at a fixed speed that (in miles per hour) is uniformly distributed between 40 and 60; each time he drives from B to A, he drives at a fixed speed that is equally likely to be either 40 or 60.

    (a) In the long run, what proportion of his driving time is spent going to B?
    (b) In the long run, for what proportion of his driving time is he driving at a speed of 40 miles per hour?

Example 7.16 on p.454-455 (p.430-431 in 11th edition), find the long-run proportion of time that the server is busy.

Example 7.16 Suppose that customers arrive at a single-server system in accordance with a Poisson process with rate λ. Upon arriving a customer must pass through a door that leads to the server. However, each time someone passes through, the door becomes locked for the next t units of time. An arrival finding a locked door is lost, and a cost c is incurred by the system. An arrival finding the door unlocked passes through to the server. If the server is free, the customer enters service; if the server is busy, the customer departs without service and a cost K is incurred. If the service time of a customer is exponential with rate μ, find the average cost per unit time incurred by the system.

Exercise 6.5 on p.420-421 (p.399 in 11ed)

5. There are N individuals in a population, some of whom have a certain infection that spreads as follows. Contacts between two members of this population occur in accordance with a Poisson process having rate λ. When a contact occurs, it is equally likely to involve any of the $\binom{N}{2}$ pairs of individuals in the population. If a contact involves an infected and a noninfected individual, then with probability p the noninfected individual becomes infected. Once infected, an individual remains infected throughout. Let $X(t)$ denote the number of infected members of the population at time t.

   (a) Is $\{X(t), t \geqslant 0\}$ a continuous-time Markov chain?
   (b) Specify its type.
   (c) Starting with a single infected individual, what is the expected time until all members are infected?

Exercise 5.15 on p.358 (p.339 in 11ed)

15. One hundred items are simultaneously put on a life test. Suppose the lifetimes of the individual items are independent exponential random variables with mean 200 hours. The test will end when there have been a total of 5 failures. If T is the time at which the test ends, find $E[T]$ and $\mathrm{Var}(T)$.

Exercise 5.20 on p.359-360 (p.341 in 11ed)

20. Consider a two-server system in which a customer is served first by server 1, then by server 2, and then departs. The service times at server i are exponential random variables with rates μ_i, $i = 1, 2$. When you arrive, you find server 1 free and two customers at server 2—customer A in service and customer B waiting in line.

   (a) Find P_A, the probability that A is still in service when you move over to server 2.
   (b) Find P_B, the probability that B is still in the system when you move over to server 2.
   (c) Find $E[T]$, where T is the time that you spend in the system.

   Hint: Write

$$T = S_1 + S_2 + W_A + W_B$$

   where S_i is your service time at server i, W_A is the amount of time you wait in queue while A is being served, and W_B is the amount of time you wait in queue while B is being served.

Problem 5.44 on p.365 (p.345 in 11ed)

44. Cars pass a certain street location according to a Poisson process with rate λ. A woman who wants to cross the street at that location waits until she can see that no cars will come by in the next T time units.

    (a) Find the probability that her waiting time is 0.
    (b) Find her expected waiting time.

Problem 5.55 on p.366 (p.347 in 11ed)

55. Consider a single server queuing system where customers arrive according to a Poisson process with rate λ, service times are exponential with rate μ, and customers are served in the order of their arrival. Suppose that a customer arrives and finds $n-1$ others in the system. Let X denote the number in the system at the moment that customer departs. Find the probability mass function of X.

Problem 5.66 on p.368 (p.349 in 11ed)

66. Policyholders of a certain insurance company have accidents at times distributed according to a Poisson process with rate λ. The amount of time from when the accident occurs until a claim is made has distribution G.

    (a) Find the probability there are exactly n incurred but as yet unreported claims at time t.
    (b) Suppose that each claim amount has distribution F, and that the claim amount is independent of the time that it takes to report the claim. Find the expected value of the sum of all incurred but as yet unreported claims at time t.